MA1522 Linear Algebra for Computing

Lesson 18: Markov Chains and Singular Value Decomposition

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson18` (e.g. `A1234567Z_lesson18.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 18**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 6.4 Powers of Diagonalizable Matrices

Theorem

If

$$A = P \begin{pmatrix} \mu_1 & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix} P^{-1},$$

then for every positive integer k,

$$A^k = P \begin{pmatrix} \mu_1^k & 0 & \cdots & 0 \\ 0 & \mu_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n^k \end{pmatrix} P^{-1}.$$

If A is invertible, this holds for every integer $k \in \mathbb{Z}$.

Section 6.4 Markov Chains

Definition

1. A vector $\mathbf{v} = (v_i) \in \mathbb{R}^n$ with nonnegative coordinates that add up to 1 is a **probability vector**.
2. A **stochastic matrix** is a square matrix whose columns are probability vectors.
3. A **Markov chain** is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k, \ldots$ together with a stochastic matrix A such that

$$\mathbf{x}_1 = A\mathbf{x}_0, \quad \mathbf{x}_2 = A\mathbf{x}_1, \quad \ldots, \quad \mathbf{x}_k = A\mathbf{x}_{k-1}, \quad \ldots$$

   The vector $\mathbf{x}_k$ is called a **state vector**, and $\mathbf{x}_k = A^k\mathbf{x}_0$.
4. An **equilibrium vector** is a probability vector $\mathbf{v}$ such that

$$A\mathbf{v} = \mathbf{v}.$$

   Namely, an eigenvector of A of eigenvalue 1.

Question 1

A bike-sharing scheme has 600 bikes and three docking stations A, B, C. Each day, every bike is ridden once and returned to a station:

- a bike taken from A is returned to B with probability 0.6 and to C with probability 0.4;
- a bike taken from B is returned to B with probability 0.8 and to C with probability 0.2;
- a bike taken from C is returned to A with probability 0.5 and to C with probability 0.5.

Today there are 300 bikes at A, 300 at B and none at C. Let $\mathbf{x}_k = (a_k, b_k, c_k)^T$ be the expected numbers of bikes at A, B, C after k days, so $\mathbf{x}_0 = (300, 300, 0)^T$.

1. Find the stochastic matrix A such that $\mathbf{x}_{k+1} = A\mathbf{x}_k$, and find $\mathbf{x}_1$ and $\mathbf{x}_2$.
2. Given the diagonalization of A on page 8, find the number of bikes at each station in the long run.

Question 1 — solution: part 1

1. The j-th column lists where a bike from the j-th station goes:

$$A = \begin{pmatrix} 0 & 0 & 0.5 \\ 0.6 & 0.8 & 0 \\ 0.4 & 0.2 & 0.5 \end{pmatrix}. \qquad \text{(each column sums to 1 } \checkmark\text{)}$$

Multiply by A, one day at a time:

$$\mathbf{x}_1 = A \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 420 \\ 180 \end{pmatrix}, \qquad \mathbf{x}_2 = A \begin{pmatrix} 0 \\ 420 \\ 180 \end{pmatrix} = \begin{pmatrix} 90 \\ 336 \\ 174 \end{pmatrix}.$$

Each still adds up to 600. $\checkmark$

Question 1 — part 2

2. The matrix A of Question 1 is diagonalized by

$$A = PDP^{-1}, \quad P = \begin{pmatrix} 1 & 1 & 5 \\ 3 & -2 & -3 \\ 2 & 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{5} \end{pmatrix}, \quad P^{-1} = \frac{1}{42} \begin{pmatrix} 7 & 7 & 7 \\ 0 & -12 & 18 \\ 7 & 1 & -5 \end{pmatrix}.$$

$\mathbf{x}_k = A^k \mathbf{x}_0 = PD^k P^{-1} \mathbf{x}_0$, and as $k \to \infty$, $(\frac{1}{2})^k \to 0$ and $(-\frac{1}{5})^k \to 0$, so $D^k \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Multiplying out from the left,

$$\mathbf{x}_k \longrightarrow \begin{pmatrix} 1 & 1 & 5 \\ 3 & -2 & -3 \\ 2 & 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{42} \begin{pmatrix} 7 & 7 & 7 \\ 0 & -12 & 18 \\ 7 & 1 & -5 \end{pmatrix} \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \frac{1}{42} \begin{pmatrix} 7 & 7 & 7 \\ 0 & -12 & 18 \\ 7 & 1 & -5 \end{pmatrix} \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 \\ 3 & 3 & 3 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 300 \\ 300 \\ 0 \end{pmatrix} = \begin{pmatrix} 100 \\ 300 \\ 200 \end{pmatrix}.$$

In the long run there are 100 bikes at A, 300 at B and 200 at C.

Section 6.4 Regular Stochastic Matrices

Definition

A stochastic matrix A is **regular** if A^k has all entries positive for some positive integer k.

Theorem

If A is a regular stochastic matrix, then A has a unique equilibrium vector $\mathbf{v}$, and for every probability vector $\mathbf{x}_0$, the Markov chain $\mathbf{x}_k = A^k\mathbf{x}_0$ converges to $\mathbf{v}$.

Algorithm to compute an equilibrium vector.

1. Find a nontrivial solution $\mathbf{u} = (u_i)$ of $(I - A)\mathbf{x} = \mathbf{0}$.
2. Then $\mathbf{v} = \dfrac{1}{u_1 + u_2 + \cdots + u_n}\,\mathbf{u}$ is an equilibrium vector.

Question 2

Let A be the matrix in Question 1,

$$A = \begin{pmatrix} 0 & 0 & 0.5 \\ 0.6 & 0.8 & 0 \\ 0.4 & 0.2 & 0.5 \end{pmatrix}.$$

1. Show that A is regular.

2. Find the equilibrium vector of A.

Question 2 — solution

1. A has zero entries, but

$$A^2 = \frac{1}{100}\begin{pmatrix} 20 & 10 & 25 \\ 48 & 64 & 30 \\ 32 & 26 & 45 \end{pmatrix}$$

has all entries positive, so A is regular.

2. Solve $(I - A)\mathbf{x} = \mathbf{0}$, using $10(I - A)$ to clear the decimals:

$$10(I - A) = \begin{pmatrix} 10 & 0 & -5 \\ -6 & 2 & 0 \\ -4 & -2 & 5 \end{pmatrix} : \quad x_3 = 2x_1, \quad x_2 = 3x_1, \qquad \mathbf{u} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

Divide by the sum $1 + 3 + 2 = 6$:

$$\mathbf{v} = \frac{1}{6}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

□

Challenge

The **Fibonacci numbers** are defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \geq 0),$$

so the sequence is $0, 1, 1, 2, 3, 5, 8, 13, \ldots$. Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$.

1. Show that $A \begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix} = \begin{pmatrix} F_{n+1} \\ F_{n+2} \end{pmatrix}$ for every $n \geq 0$.
2. Diagonalize A, and show that for every $n \geq 0$,

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right].$$

Hint: by part 1, $A^n \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix}$.

Section 6.5 Singular Value Decomposition

Definition

Let A be an $m \times n$ matrix. A **singular value decomposition** of A is an expression

$$A = U\Sigma V^T,$$

where

- U and V are $m \times m$ and $n \times n$ orthogonal matrices;
- Σ is an $m \times n$ "diagonal matrix":

$$\Sigma = \begin{pmatrix} D & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)} \end{pmatrix}, \qquad D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r), \quad \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Theorem

Every $m \times n$ matrix A has a singular value decomposition. Here $r = \operatorname{rank}(A)$, and $\sigma_1 \geq \cdots \geq \sigma_n$ are square roots of the eigenvalues of $A^T A$ (the **singular values** of A).

Section 6.5 Finding a Singular Value Decomposition

Suppose that $A = U\Sigma V^T$ is a singular value decomposition.

- $A^TA = V\Sigma^T\Sigma V^T$, with $\Sigma^T\Sigma = \text{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)$. So V orthogonally diagonalizes the symmetric matrix A^TA. This determines Σ (with $\sigma_1 \geq \cdots \geq \sigma_r > 0$) and a choice of V.

- Write $U = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r & \mathbf{u}_{r+1} & \cdots & \mathbf{u}_m \end{pmatrix}$. Then

$$AV = U\Sigma = \begin{pmatrix} \sigma_1\mathbf{u}_1 & \cdots & \sigma_r\mathbf{u}_r & \mathbf{0} & \cdots & \mathbf{0} \end{pmatrix}.$$

  This determines $\mathbf{u}_i = \frac{1}{\sigma_i}A\mathbf{v}_i$, for $1 \leq i \leq r$.

- Extend $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ to an orthonormal basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ of $\mathbb{R}^m$. This determines U.

Section 6.5 Algorithm for Singular Value Decomposition

Let A be an $m \times n$ matrix of rank r.

1. Find the eigenvalues $\mu_1 \geq \cdots \geq \mu_r > 0 = \mu_{r+1} = \cdots = \mu_n$ of $A^T A$; let $\sigma_i = \sqrt{\mu_i}$ and form Σ.
2. Find an orthonormal basis of each eigenspace of $A^T A$; let $\mathbf{v}_i$ be the unit eigenvector for μ_i, and $V = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix}$.
3. Let $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ for $i = 1, \ldots, r$. Extend to an orthonormal basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ of $\mathbb{R}^m$ by solving $\begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{pmatrix}^T \mathbf{x} = \mathbf{0}$, and let $U = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_m \end{pmatrix}$.

Exercise.

1. Show that the eigenvalues of the symmetric matrix $A^T A$ are nonnegative.
2. Show that in step 3, $\{A\mathbf{v}_1, \ldots, A\mathbf{v}_r\}$ is orthogonal.

Question 3

Find a singular value decomposition of

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -2 & 0 \end{pmatrix}.$$

That is, find orthogonal matrices U (of order 3) and V (of order 2), and the 3×2 matrix Σ, such that $A = U\Sigma V^T$.

Question 3 — solution: Σ and V

Orthogonally diagonalize A^TA:

$$A^TA = \begin{pmatrix} 8 & 2 \\ 2 & 5 \end{pmatrix}, \qquad \det(xI - A^TA) = (x-8)(x-5) - 4 = (x-9)(x-4).$$

So $\sigma_1 = 3$, $\sigma_2 = 2$, and

$$\Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

$$9I - A^TA = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} : \ \mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad 4I - A^TA = \begin{pmatrix} -4 & -2 \\ -2 & -1 \end{pmatrix} : \ \mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

$$V = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

Question 3 — solution: U

Let $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$:

$$\mathbf{u}_1 = \frac{1}{3}A\mathbf{v}_1 = \frac{1}{3\sqrt{5}}\begin{pmatrix}5\\2\\-4\end{pmatrix}, \qquad \mathbf{u}_2 = \frac{1}{2}A\mathbf{v}_2 = \frac{1}{2\sqrt{5}}\begin{pmatrix}0\\-4\\-2\end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix}0\\-2\\-1\end{pmatrix}.$$

Extend to $\mathbb{R}^3$: apply Gram–Schmidt to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, where $\mathbf{e}_1 = (1,0,0)^T$, $\mathbf{e}_2 = (0,1,0)^T$, $\mathbf{e}_3 = (0,0,1)^T$. Since $\mathbf{e}_1 \cdot \mathbf{u}_1 = \frac{\sqrt{5}}{3}$ and $\mathbf{e}_1 \cdot \mathbf{u}_2 = 0$,

$$\mathbf{e}_1 - \tfrac{\sqrt{5}}{3}\mathbf{u}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix} - \frac{1}{9}\begin{pmatrix}5\\2\\-4\end{pmatrix} = \frac{1}{9}\begin{pmatrix}4\\-2\\4\end{pmatrix}, \qquad \mathbf{u}_3 = \frac{1}{3}\begin{pmatrix}2\\-1\\2\end{pmatrix}.$$

Then $\mathbf{e}_2$ and $\mathbf{e}_3$ give $\mathbf{0}$ (they lie in $\operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \mathbb{R}^3$) and are discarded.

$$U = \begin{pmatrix}5/(3\sqrt{5}) & 0 & 2/3\\ 2/(3\sqrt{5}) & -2/\sqrt{5} & -1/3\\ -4/(3\sqrt{5}) & -1/\sqrt{5} & 2/3\end{pmatrix}, \qquad A = U\Sigma V^T. \qquad \square$$

Homework

1. Search for applications of the singular value decomposition.

2. Find a singular value decomposition of each of the following matrices.

$$H_1 = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 2 & 1 \\ 2 & 1 \\ 2 & 1 \end{pmatrix}.$$

Next lesson

Next lesson (**Fri 30 Oct**, Section 7.1), I will focus on

1. linear transformations, and

2. the standard matrix of a linear transformation.

Please watch the video clips beforehand.

Exercise (slide 15) — solution

1. Let $A^T A\mathbf{v} = \mu\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then

$$\mu\|\mathbf{v}\|^2 = \mathbf{v}^T(\mu\mathbf{v}) = \mathbf{v}^T A^T A\mathbf{v} = (A\mathbf{v})^T(A\mathbf{v}) = \|A\mathbf{v}\|^2 \geq 0,$$

and $\|\mathbf{v}\|^2 > 0$, so $\mu \geq 0$.

2. The $\mathbf{v}_i$ are orthonormal eigenvectors of $A^T A$, with $A^T A\mathbf{v}_j = \mu_j\mathbf{v}_j$. For $i \neq j$,

$$(A\mathbf{v}_i) \cdot (A\mathbf{v}_j) = \mathbf{v}_i^T A^T A\mathbf{v}_j = \mu_j\, \mathbf{v}_i \cdot \mathbf{v}_j = 0.$$

So $\{A\mathbf{v}_1, \ldots, A\mathbf{v}_r\}$ is orthogonal. (With $i = j$: $\|A\mathbf{v}_i\|^2 = \mu_i = \sigma_i^2$, so each $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ is a unit vector.) □

Homework — answers

$$H_1 = \frac{1}{\sqrt{10}}\begin{pmatrix}1 & 3\\ 3 & -1\end{pmatrix}\begin{pmatrix}3\sqrt{5} & 0\\ 0 & \sqrt{5}\end{pmatrix}\left[\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\right]^T, \quad H_2 = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}2\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\left[\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\right]^T,$$

$$H_3 = \begin{pmatrix}1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3}\\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3}\\ 2/\sqrt{6} & 0 & -1/\sqrt{3}\end{pmatrix}\begin{pmatrix}\sqrt{6} & 0\\ 0 & \sqrt{2}\\ 0 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}^T,$$

$$H_4 = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}\sqrt{3} & 0 & 0\\ 0 & 1 & 0\end{pmatrix}\begin{pmatrix}1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3}\\ 2/\sqrt{6} & 0 & -1/\sqrt{3}\\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3}\end{pmatrix}^T,$$

$$H_5 = \begin{pmatrix}1/\sqrt{3} & 2/\sqrt{6} & 0\\ 1/\sqrt{3} & -1/\sqrt{6} & 1/\sqrt{2}\\ 1/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{2}\end{pmatrix}\begin{pmatrix}\sqrt{15} & 0\\ 0 & 0\\ 0 & 0\end{pmatrix}\left[\frac{1}{\sqrt{5}}\begin{pmatrix}2 & 1\\ 1 & -2\end{pmatrix}\right]^T.$$

Challenge — solution

1. $A\begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix} = \begin{pmatrix} F_{n+1} \\ F_n + F_{n+1} \end{pmatrix} = \begin{pmatrix} F_{n+1} \\ F_{n+2} \end{pmatrix}$. Applying this n times to $(F_0, F_1)^T = (0,1)^T$ gives $A^n(0,1)^T = (F_n, F_{n+1})^T$.

2. $\det(xI - A) = x^2 - x - 1$, with roots $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. For each root λ, $A(1,\lambda)^T = (\lambda, 1+\lambda)^T = \lambda(1,\lambda)^T$, since $\lambda^2 = \lambda + 1$. So

$$P = \begin{pmatrix} 1 & 1 \\ \varphi & \psi \end{pmatrix}, \quad D = \begin{pmatrix} \varphi & 0 \\ 0 & \psi \end{pmatrix}, \quad P^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\psi - \varphi}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix} = PD^nP^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 & 1 \\ \varphi & \psi \end{pmatrix}\begin{pmatrix} \varphi^n \\ -\psi^n \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} \varphi^n - \psi^n \\ \varphi^{n+1} - \psi^{n+1} \end{pmatrix}.$$

The first entry is the formula for F_n. □